

Precision Policy Optimisation: A Fully Agnostic Causal Machine Learning Framework for Resource Allocation Under Constraint

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Abstract

Traditional approaches to public policy evaluation often rely on average treatment effect estimation, which inherently masks the presence of heterogeneous treatment effects and fails to provide reasonable, decision-level guidance. This paper presents a generalised, fully agnostic Causal Machine Learning (Causal Forest) pipeline designed to optimise resource allocation in resource-constrained environments. By shifting from standard randomised control trial (RCT) methodologies to a model that explicitly accounts for latent propensity scores and selection bias, this framework enables public sector agencies to transition from reactive implementation to predictive "Active-Lift" strategies.

I stress-test the architectural robustness of this pipeline across ten distinct, simulated policy domains, ranging from educational interventions and telehealth patient monitoring to urban wastewater infrastructure and small business digital transformation. Each scenario is calibrated with domain-specific budget capacities (varying from 10% to 35%) and unique selection biases, simulating the complex reality of real-world administrative data.

The findings demonstrate that by integrating the Causal Forest (GRF) methodology with a modular "Fiscal Risk Exposure" engine, administrators can quantify and minimise the allocation of capital to non-responsive or low-impact applicants. The framework introduces an "Executive Summary Canvas" translation layer, which synthesises high-dimensional causal outcomes into intuitive, operational policy briefings. This research establishes a roadmap for evidence-based governance, offering a scalable method to maximise social return on investment (ROI) while maintaining strict fiscal discipline in the face of heterogeneous population needs.

1. Introduction

In the contemporary landscape of public administration, policymakers are increasingly confronted by a challenging paradox: the demand for bespoke, high-impact social interventions is rising, yet the fiscal environments required to sustain them are becoming increasingly restricted. Historically, evidence-based policy has been governed by the "average treatment effect" (ATE) paradigm, derived from randomised control trials (RCTs). While robust for determining whether an intervention works on average, this aggregate focus frequently obscures the nuance of heterogeneous treatment effects, the reality that interventions may yield significant positive outcomes for certain sub-groups while remaining ineffective, or even detrimental, for others.

The reliance on ATE-based decision-making often leads to the inefficient allocation of public capital. When a policy is deployed as a one-size-fits-all measure, substantial resources are inadvertently expended on non-responsive applicants, while those with the highest potential for impact remain underserved. This systemic fiscal leakage is not merely an accounting concern; it represents an opportunity cost where social return on investment (ROI) is systematically minimised. As digital transformation continues to integrate administrative data across diverse sectors, from education and healthcare to urban infrastructure, the technical capacity to move beyond aggregate analysis has finally caught up with the imperative to do so.

This paper introduces a fully agnostic, self-validating Causal Machine Learning pipeline underpinned by the Generalised Random Forest (GRF) methodology (Athey, Tibshirani, & Wager, 2019). By pivoting from predictive modelling to causal inference, this framework enables public sector agencies to transition from reactive, intuition-based programme management to predictive *Active-Lift* strategies. Unlike traditional regression models, which can be vulnerable to selection bias and omitted variable sensitivity, GRF architectures offer a non-parametric method for estimating conditional average

treatment effects (CATE) with asymptotic properties that allow for valid statistical inference (Wager & Athey, 2018). Furthermore, the application of these techniques to resource allocation draws upon recent advancements in policy learning, providing a robust pathway for leveraging observational data to optimise social returns (Athey & Wager, 2021).

The central innovation of this approach lies in the integration of a budget capacity constraint layer within the causal inference loop. By defining strict resource boundaries, the maximum percentage of the applicant pool that can receive a treatment, the algorithm is trained not merely to predict outcomes, but to optimise policy assignment to those individuals whose predicted uplift is highest. This effectively turns the constraint into an optimisation target, allowing administrators to quantify the fiscal risk exposure of any given policy strategy.

To stress-test this architecture, the research employs a multi-domain validation framework, applying the GRF pipeline to ten distinct policy scenarios. These scenarios span a diverse range of operational contexts: from educational interventions and telehealth patient monitoring to urban wastewater infrastructure and small business digital transformation. By synthesising these varying domains, we demonstrate the scalability of this pipeline, providing a replicable roadmap for evidence-based governance that maximises social welfare whilst maintaining rigorous fiscal discipline.

2. Foundational Framework: The Neyman-Rubin Causal Model

Before diving into the algorithms, we must understand the structural framework under which these estimators operate. The pipeline handles data consisting of triplets (X_i, Y_i, D_i) for individuals $i = 1, \dots, N$:

- $X_i \in X \subset \mathbb{R}^p$: A multi-dimensional feature vector (covariates).
- $D_i \in \{0,1\}$: A binary treatment/policy indicator variable.
- $Y_i \in \mathbb{R}$: The observed continuous scalar outcome metric.

2.1. Potential Outcomes & The Fundamental Problem

Following Rubin's formulation, every individual possesses two latent **Potential Outcomes**:

- $Y_i(1)$: The outcome if treated ($D_i = 1$).
- $Y_i(0)$: The outcome if untreated ($D_i = 0$).

The observable outcome is tied to reality via the switching equation:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

Because we can never observe both potential outcomes simultaneously for a single individual i , the individual treatment effect $Y_i(1) - Y_i(0)$ is fundamentally unobservable.

2.2. The Target Parameter: CATE

Instead, the machine learning architecture targets the **Conditional Average Treatment Effect (CATE)**, denoted as $\tau(x)$:

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$$

2.3. Identifiability Assumptions

To guarantee that $\tau(x)$ reflects a true causal link rather than a spurious correlation, three standard axioms must hold within the population data stream:

1. **Unconfoundedness (Conditional Independence):** $\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i$. Given the measured covariates, treatment assignment is as good as random.
2. **Overlap (Positivity):** $0 < \epsilon \leq P(D_i = 1 \mid X_i = x) \leq 1 - \epsilon < 1$ for all $x \in X$. Every individual has a non-zero probability of receiving or being denied treatment.
3. **SUTVA (Stable Unit Treatment Value Assumption):** The potential outcomes of any individual are unaffected by the treatment assignments of other individuals, and there are no hidden variations in the delivery of the treatment.

3. Algorithm 1: Causal Random Forest (CRF)

3.1. Aim

The overarching aim of the Causal Random Forest (Wager & Athey, 2018) is to non-parametrically isolate and estimate the heterogeneous treatment effect surface $\tau(x)$ over a high-dimensional covariate space X , without imposing rigid functional assumptions (such as linearity) on the baseline outcome or the treatment effects.

3.2. Step A: Robinson's Orthogonalisation (Targeted De-biasing)

A standard machine learning model trying to predict Y from X and D will become severely biased if X affects both the assignment of D (confounding) and the scale of Y (main effects). To completely isolate the pure treatment effect, CRF implements **Robinson's (1988) Double-Residualisation**:

1. Estimate the **Conditional Mean Outcome**: $m(x) = E[Y \mid X = x]$
2. Estimate the **Propensity Score**: $e(x) = P(D = 1 \mid X = x)$

In the codes, `p_forest` acts as the engine calculating $\hat{e}(x)$. Using these estimates, the data is transformed into centered residuals:

$$\begin{aligned}\tilde{Y}_i &= Y_i - \hat{m}(X_i) \\ \tilde{D}_i &= D_i - \hat{e}(X_i)\end{aligned}$$

By subtracting the baseline expectations, we can state a clean orthogonalized regression problem:

$$\tilde{Y}_i = \tau(X_i)\tilde{D}_i + \epsilon_i, \quad E[\epsilon_i \mid X_i] = 0$$

3.3. Step B: Local Estimating Equations & Generalized Random Forests (GRF)

Instead of minimising Mean Squared Error like a classic random forest, a Causal Forest builds upon the Generalized Random Forest (GRF) framework (Athey, Tibshirani, & Wager, 2019). It turns the forest into an adaptive kernel estimator that solves a localized version of an estimating equation.

For a target point x , the forest computes a set of adaptive weights $\alpha_i(x)$. These weights quantify how often observation i falls into the same terminal leaf node as the target point x across all B trees in the forest:

$$\alpha_i(x) = \frac{1}{B} \sum_{b=1}^B \frac{\mathbb{1}(\{X_i \in L_b(x)\})}{|L_b(x)|}$$

Where $L_b(x)$ denotes the terminal leaf of the b -th tree containing point x , and $|L_b(x)|$ is the number of training samples located within that leaf.

The Causal Forest then evaluates the local optimisation problem to solve for $\hat{\tau}(x)$:

$$\hat{\tau}(x) = \arg \min_{\tau} \left\{ \sum_{i=1}^N \alpha_i(x) (\tilde{Y}_i - \tau \cdot \tilde{D}_i)^2 \right\}$$

Differentiating with respect to τ yields the explicit local Wald estimator used in the code:

$$\hat{\tau}(x) = \frac{\sum_{i=1}^N \alpha_i(x) \tilde{D}_i \tilde{Y}_i}{\sum_{i=1}^N \alpha_i(x) \tilde{D}_i^2}$$

3.4. Step C: Split Criterion (Maximising Treatment Heterogeneity)

When growing each tree, the algorithm cannot use standard regression splits because $\tau(X_i)$ is unknown at the individual level. Instead, when considering splitting a parent node P into a Left (L) and Right (R) child node, it selects the split feature and cutoff value that maximizes the targeted variance of the treatment effects across the children:

$$\Delta(L, R) = \frac{N_L N_R}{N_P^2} (\hat{\tau}_L - \hat{\tau}_R)^2$$

Where N_L, N_R are the sample counts in each node, and $\hat{\tau}_L, \hat{\tau}_R$ are the mean treatment effects computed within those prospective splits.

3.5. Step D: Honest Tree Inference

To ensure standard errors (CATE_SE) are statistically valid, the forest enforces **Honesty**.

For each individual tree b :

1. The subsample of training data chosen for that tree is split exactly in half into two distinct pools: Subsample $\mathcal{I}_{split}$ and Subsample $\mathcal{I}_{estimate}$.
2. $\mathcal{I}_{split}$ is used exclusively to determine the structural geometry of the splits (the features and thresholds).
3. $\mathcal{I}_{estimate}$ is used exclusively to compute the final local parameters ($\hat{\tau}$) within the established leaves.

Pedagogical Note: Honesty completely uncouples the model's structural selection bias from its parameter estimation. This forces the bias of the estimators to decay asymptotically, enabling the application of the **Infinitesimal Jackknife (IJ)** to construct valid Gaussian confidence intervals around the causal predictions.

4. The Bridge: Augmented Inverse Probability Weighting (AIPW)

Before transitioning to the policy tree, the script generates `dr_scores` using a technique called Double-Robust Scoring. This step bridges treatment effect estimation and policy optimisation.

4.1. Aim

The aim is to build an unbiased, person-by-person proxy of the potential outcomes $\Gamma_i(1)$ and $\Gamma_i(0)$ that remains robust even if either the outcome model or the propensity model is misspecified.

4.2. Mathematical Formulation

The vector scores are constructed via Augmented Inverse Probability Weighting (AIPW):

$$\Gamma_i(1) = \hat{\mu}_1(X_i) + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)}$$

$$\Gamma_i(0) = \hat{\mu}_0(X_i) + \frac{(1 - D_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)}$$

Where $\hat{\mu}_1(x) = E[Y | X = x, D = 1]$ and $\hat{\mu}_0(x) = E[Y | X = x, D = 0]$ represent the conditional outcome surfaces extracted implicitly from the forest architecture.

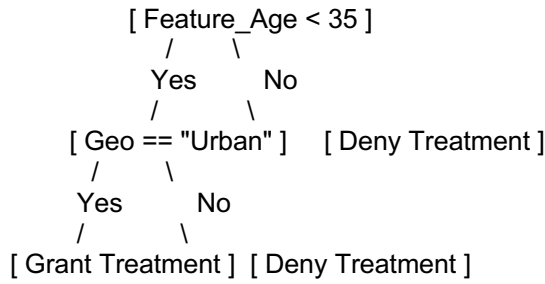
In Step 5, the policy engine transforms these into an operational reward matrix, injecting the program cost parameter c (program_cost) directly into the treated channel:

$$\text{Reward Matrix} = [\Gamma_i(0), \Gamma_i(1) - c]$$

5. Algorithm 2: Optimal Policy Tree

5.1. Aim

The aim of the Policy Tree algorithm (Athey & Wager, 2021) is to find a simple, deterministic, and easily interpretable rule mapping $\pi(x): X \rightarrow \{0,1\}$ that maximizes total societal benefit or public welfare, while operating under physical resource boundaries.



5.2. Step A: Empirical Welfare Maximisation (EWM)

Let $\pi \in \Pi$ represent a policy rule within a class of decision trees Π of fixed depth L (set to depth = 2 in the script). The empirical value mapping $Q(\pi)$ represents the estimated per-capita value achieved if policy π dictates resource distribution:

$$Q(\pi) = \frac{1}{N} \sum_{i=1}^N [\mathbb{1}(\pi(X_i) = 1)(\Gamma_i(1) - c) + \mathbb{1}(\pi(X_i) = 0)\Gamma_i(0)]$$

The algorithm searches for the optimal policy mapping π^* that satisfies:

$$\pi^* = \arg \max_{\pi \in \Pi} Q(\pi)$$

We can rewrite this optimisation problem relative to a baseline to see how it isolates pure net value. By subtracting the baseline value of denying treatment to everyone, the optimisation problem simplifies to finding a policy that maximizes the net gains:

$$\pi^* = \arg \max_{\pi \in \Pi} \left\{ \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\pi(X_i) = 1) \cdot \Delta_i \right\}$$

Where $\Delta_i = \Gamma_i(1) - \Gamma_i(0) - c$ represents the net double-robust individual lift minus institutional cost.

5.3. Step B: Global Non-Greedy Tree Search Optimisation

Standard tree classifiers like CART (e.g., `rpart`) build trees using a **greedy, top-down strategy**. They look for the single best split at the root node, freeze it, and then move down to the next level. This approach often misses the globally optimal strategy when multiple variables interact in complex ways.

In contrast, the `policytree` algorithm performs a **global exact search** for a shallow tree of depth L :

1. It evaluates the joint distribution of all combinations of splits across all L levels simultaneously.
2. It uses dynamic programming along with fast matrix operations to prune away suboptimal execution branches early.
3. This exhaustive search guarantees that the resulting tree achieves the absolute maximum empirical welfare $Q(\pi)$ across the entire structural dictionary of depth L .

Table 1. Summary checklist of the accompanying code

STRUCTURAL STAGE	APPLIED MATHEMATICAL ROUTINE	COMPUTATIONAL OBJECTIVE
STEP 4: CAUSAL FOREST	Robinson Orthogonalisation & Local Wald Estimating Equations	Minimises confounding bias to isolate continuous, individual-level CATE estimates.
STEP 5: BRIDGE	Augmented Inverse Probability Weighting (AIPW)	Builds unbiased counterfactual proxies ($\Gamma(0), \Gamma(1)$) that remain robust to model misspecification.
STEP 5: POLICY TREE	Global Empirical Welfare Maximisation (EWM)	Solves a global optimisation problem to extract a highly interpretable, non-greedy decision tree for resource allocation.

6. Policy Simulation Scenarios

This section outlines the true data generation processes for ten distinct policy simulations. In real-world data, the exact functional forms of selection bias (propensity) and impact (treatment effect) are unknown. These synthetic environments explicitly define these mechanisms so that the performance of the Generalized Random Forest (GRF) and policy tree algorithms can be scientifically validated.

In all ten scenarios, the treatment/policy assignment D is generated stochastically for each individual or unit based on their calculated propensity score $p(x)$.

Mathematically, the treatment D is drawn from a Bernoulli distribution, where the probability of receiving the treatment ($D = 1$) is exactly equal to the propensity score given the observed covariates $X = x$:

$$D \sim \text{Bernoulli}(p(x))$$

Which is equivalent to stating:

$$P(D = 1 \mid X = x) = p(x)$$

Below are the explicit logistic equations used to calculate the propensity score $p(x)$ for each of the ten policy scenarios. The indicator function $I(\cdot)$ evaluates to 1 if the condition is met and 0 otherwise.

6.1. After-School Intensive Tutoring

Policy Objective: Targeted after-school mathematics tutoring designed to improve standardised test scores.

Sample Size (N): 2000

Outcome Variable (Metric_Y): Standardised test score (capped at a maximum of 100).

Covariates:

- Feature_GradeLevel: Student's grade level (Integer: 6–12).
- Feature_SES: Socioeconomic status (Categorical: Low, Mid, High).
- Feature_PriorScore: Previous test performance ($\sim \mathcal{N}(50, 15^2)$).
- Feature_Attendance: Baseline attendance rate (Continuous: 0.6–1.0).

Data Generation Mechanics:

- **Selection Bias (Propensity):** School counsellors actively target low-SES students and those with lower prior scores.

$$p(x) = \frac{1}{1 + \exp(-(-0.5 + 2 \cdot I(\text{SES} = \text{Low}) - 0.05 \cdot \text{PriorScore}))}$$

- **True Treatment Effect (τ):** The tutoring provides a baseline lift, scaling with prior knowledge, but highly benefits low-SES students.

$$\tau(x) = 10 + 0.2 \cdot \text{PriorScore} + 5 \cdot I(\text{SES} = \text{Low})$$

- **Observed Outcome:**

$$Y = \min(100, 40 + 0.8 \cdot \text{PriorScore} + D \cdot \tau(x) + \epsilon)$$

$$\epsilon \sim \mathcal{N}(0, 5^2)$$

6.2. Remote Patient Monitoring (Telemedicine)

Policy Objective: Providing Remote Patient Monitoring (RPM) equipment to manage chronic diseases and reduce hospital readmissions.

Sample Size (N): 2500

Outcome Variable (Metric_Y): Total 30-day hospital readmission costs (in dollars). The treatment aims to *reduce* this value.

Covariates:

- Feature_Age: Patient age ($\sim \mathcal{N}(65, 10^2)$).
- Feature_ComorbidtyCount: Number of co-existing conditions (Poisson, $\lambda = 2$).
- Feature_DigitalLiteracy: Patient's tech proficiency (Categorical: Low, Medium, High).
- Feature_DistanceToClinic: Miles to the nearest facility (Continuous: 5–100).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Patients with more comorbidities are triaged into the program, but proximity to the clinic increases likelihood of enrolment.

$$p(x) = \frac{1}{1 + \exp(-(-1.5 + 0.1 \cdot \text{Comorbidities} - 0.02 \cdot \text{Distance}))}$$

- **True Treatment Effect (τ):** RPM avoids significant costs for highly comorbid patients and those with low digital literacy (who otherwise mismanage medications).

$$\tau(x) = 1500 + 500 \cdot \text{Comorbidities} + 1000 \cdot I(\text{Literacy} = \text{Low})$$

- **Observed Outcome:**

$$Y = \max(0, 5000 + 1000 \cdot \text{Comorbidities} + 50 \cdot \text{Age} - D \cdot \tau(x) + \epsilon)$$

$$\epsilon \sim \mathcal{N}(0, 500^2)$$

6.3. Residential Energy Efficiency Retrofit

Policy Objective: Municipal grants for home insulation and heat pump installations to lower carbon footprints.

Sample Size (N): 2000

Outcome Variable (Metric_Y): Annual residential energy bill (in dollars). The treatment aims to *reduce* this value.

Covariates:

- Feature_HouseAge: Age of the property in years (Continuous: 5–120).
- Feature_HeatingType: Primary heat source (Categorical: Electric, Gas, Oil).
- Feature_InsulationLevel: Current structural efficiency (Categorical: Poor, Average, Good).
- Feature_EnergyUsage: Baseline kWh usage ($\sim \mathcal{N}(15000, 3000^2)$).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Older homes are systematically prioritised for the grant program.

$$p(x) = \frac{1}{1 + \exp(-(-2 + 0.05 \cdot (120 - \text{HouseAge})))}$$

- **True Treatment Effect (τ):** Retrofits scale directly with prior energy usage and are exceptionally effective at replacing inefficient electric heating.

$$\tau(x) = 1000 + 0.2 \cdot \text{EnergyUsage} + 500 \cdot I(\text{Heating} = \text{Electric})$$

- **Observed Outcome:**

$$Y = 2000 - D \cdot \tau(x) + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, 200^2)$$

6.4. Small Business Digital Transformation

Policy Objective: Government grants subsidising consulting and software adoption to spur local economic growth.

Sample Size (N): 1500

Outcome Variable (Metric_Y): Annual business revenue growth (in dollars).

Covariates:

- Feature_BusinessAge: Years in operation (Continuous: 1–40).
- Feature_Sector: Industry type (Categorical: Retail, Manufacturing, Services).
- Feature_NumEmployees: Workforce size (Poisson, $\lambda = 10$).
- Feature_CurrentTechLevel: Pre-intervention digitisation (Categorical: Low, Medium, High).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Larger businesses possess the administrative capacity to successfully apply for grants, increasing their selection probability as their workforce grows.

$$p(x) = \frac{1}{1 + \exp(-(-1.5 + 0.1 \cdot \text{Employees}))}$$

- **True Treatment Effect (τ):** The grant scales massively with workforce size and provides the largest jump to technologically lagging firms.

$$\tau(x) = 5000 + 1000 \cdot \text{Employees} + 3000 \cdot I(\text{TechLevel} = \text{Low})$$

- **Observed Outcome:**

$$Y = 50000 + 500 \cdot \text{BusinessAge} + D \cdot \tau(x) + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, 5000^2)$$

6.5. Smart Irrigation Subsidy

Policy Objective: Subsidising agricultural smart sensors to optimise water usage against climate change.

Sample Size (N): 1500

Outcome Variable (Metric_Y): Crop yield (kilograms per hectare).

Covariates:

- Feature_SoilQuality: Agronomic baseline metric (Continuous: 0–100).
- Feature_CropType: Cultivated product (Categorical: Corn, Soy, Wheat).
- Feature_FarmSize: Land mass in hectares (Continuous: 10–500).
- Feature_WaterStress: Environmental drought index (Continuous: 0.0–1.0).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Large commercial farms in highly water-stressed regions aggressively capture the subsidies.

$$p(x) = \frac{1}{1 + \exp(-(-2 + 0.01 \cdot \text{FarmSize} + 0.5 \cdot \text{WaterStress}))}$$

- **True Treatment Effect (τ):** Sensors provide the highest marginal yield returns in severe drought conditions and rich soils.

$$\tau(x) = 50 + 100 \cdot \text{WaterStress} + 0.5 \cdot \text{SoilQuality}$$

- **Observed Outcome:**

$$Y = 2000 + 10 \cdot \text{SoilQuality} + D \cdot \tau(x) + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, 100^2)$$

6.6. Academic Staff Incentives (Pilot Study)

Policy Objective: Evaluating the causal return of providing financial seed capital to academic researchers.

Sample Size (N): 2000

Outcome Variable (Metric_Y): Total academic productivity (Publication counts weighted by Impact Factor).

Covariates:

- Feature_Rank: Career stage (Categorical: Postdoc to Full Professor).
- Feature_Field: Discipline (Categorical: STEM, Social Sciences, Humanities).
- Feature_Tier: Institutional prestige (Categorical: Tier 1 to Tier 3).
- Feature_YearsPhD: Career longevity, bounded realistically by academic rank.
- Feature_PriorImpact: Baseline historic productivity (Simulated via field/rank multipliers).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Structural advantage exists where Tier 1 academics and historic high-performers are more likely to win grants. To ensure strict common support for the causal forest, the probabilities are artificially bounded between 0.2 and 0.8.

$$p(x) = \max\left(0.2, \min\left(0.8, \frac{1}{1 + \exp(-(-0.2 + 0.3 \cdot I(\text{Tier } 1) - 0.2 \cdot I(\text{Tier } 3) + 0.02 \cdot \text{PriorImpact}))}\right)\right)$$

- **True Treatment Effect (τ):** A complex, non-linear conditional logic tree. Early-to-mid career scholars (Assistant/Associate) in resource-intensive fields (STEM/Social Sciences) yield the highest marginal returns. Full professors are "saturated" and yield low returns. Tier 1 amplifies the return by a factor of 1.2, while Tier 3 dampens it by 0.8.
- **Observed Outcome:**

$$Y = \max(0, 0.8 \cdot \text{PriorImpact} + 0.5 \cdot \text{YearsPhD} + \text{FieldBonus} + D \cdot \tau(x) + \epsilon)$$

$$\epsilon \sim \mathcal{N}(0, 3^2)$$

6.7. General Training Incentives (Active-Lift Template)

Policy Objective: A baseline synthetic template used for generalised system testing (e.g., job training interventions).

Sample Size (N): 1500

Outcome Variable (Metric_Y): Generalised performance or economic score.

Covariates:

- Feature_Age: Subject age (Integer: 18–72).
- Feature_Score: Prior aptitude or merit score (Continuous: 5–95).
- Feature_Geography: Location type (Categorical: Urban, Rural, Suburban).
- Feature_Tier: Stratification category (Categorical: Tier 1 to Tier 3).

Data Generation Mechanics:

- **Selection Bias (Propensity):** This scenario simulates a pure Randomized Control Trial (RCT) without confounding selection bias. Every participant has a static 45% chance of receiving the training.

$$p(x) = 0.45$$

- **True Treatment Effect (τ):** A sharp, hard-coded structural break. The intervention is highly potent for the young, urban demographic, and weak elsewhere.

$$\tau(x) = \begin{cases} 6500 & \text{if Age} < 35 \text{ and Geography} = \text{Urban} \\ 500 & \text{otherwise} \end{cases}$$

- **Observed Outcome:**

$$Y = \text{round}(22000 + 10 \cdot \text{Score} - 5 \cdot \text{Age} + D \cdot \tau(x) + \epsilon)$$

$$\epsilon \sim \mathcal{N}(0, 1000^2)$$

6.8. Urban Public Transit Subsidy

Policy Objective: Subsidising annual transit passes to reduce private vehicle commuting.

Sample Size (N): 2500

Outcome Variable (Metric_Y): Commuter carbon footprint (equivalent CO₂ emissions). The treatment aims to *reduce* this value.

Covariates:

- Feature_IncomeBracket: Commuter wealth (Categorical: Low, Mid, High).
- Feature_CommuteDistance: Miles to workplace (Continuous: 2–30).
- Feature_CurrentMode: Existing transit method (Categorical: Car, Bicycle, Walking).
- Feature_Age: Commuter age (Integer: 18–70).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Low-income commuters heavily apply for the subsidy, while long-distance commuters are slightly less likely to participate due to the limits of the transit network.

$$p(x) = \frac{1}{1 + \exp(-(-1 + 0.8 \cdot I(\text{Income} = \text{Low}) - 0.03 \cdot \text{CommuteDistance}))}$$

- **True Treatment Effect (τ):** The subsidy pulls long-distance drivers off the road and provides structural relief to low-income populations.

$$\tau(x) = 10 + 2 \cdot \text{Distance} + 5 \cdot I(\text{Income} = \text{Low})$$

- **Observed Outcome:**

$$Y = 50 - D \cdot \tau(x) + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, 5^2)$$

6.9. Urban Tree Canopy Maintenance

Policy Objective: Intensive municipal forestry maintenance to combat the urban heat island effect.

Sample Size (N): 1200

Outcome Variable (Metric_Y): Average localised neighborhood temperature (Celsius). The treatment aims to *reduce* this value.

Covariates:

- Feature_PopulationDensity: People per square mile (Continuous: 1000–20000).
- Feature_CanopyPercentage: Current tree coverage (Continuous: 5–60).
- Feature_VulnerabilityIndex: Climate risk status (Categorical: Low, Medium, High).
- Feature_ZoneSize: Size of the maintenance block (Continuous: 50–500).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Urban planners actively target zones identified as highly vulnerable to the urban heat island effect for maintenance budgets.

$$p(x) = \frac{1}{1 + \exp\left(-(-2 + 0.5 \cdot I(\text{VulnIndex} = \text{High}))\right)}$$

- **True Treatment Effect (τ):** Tree maintenance provides the best cooling returns in dense concrete environments and highly vulnerable sectors.

$$\tau(x) = 2 + 0.05 \cdot \left(\frac{\text{Density}}{1000}\right) + 3 \cdot I(\text{VulnIndex} = \text{High})$$

- **Observed Outcome:**

$$Y = 30 - D \cdot \tau(x) + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, 0.5^2)$$

6.10. Wastewater Infrastructure & Smart Grids

Policy Objective: Municipal grants to install smart overflow sensors in sewer catchments to prevent toxic spills.

Sample Size (N): 2500

Outcome Variable (Metric_Y): Annual avoided financial damages, clean-up costs, and compliance fines (in dollars).

Covariates:

- Feature_SewerType: Infrastructure design (Categorical: Combined Sewer, Separate Sewer).
- Feature_Catchment: Geographic layout (Categorical: Urban Core, Suburban, Industrial).
- Feature_AssetAge: Age of underground pipes in years (Continuous: 15–95).
- Feature_PriorSpills: Historic failure rate (Poisson, conditionally dependent on Age and Type).

Data Generation Mechanics:

- **Selection Bias (Propensity):** Zones with older underground assets and worse historic spill records successfully lobby the municipality for intervention. This is bounded to ensure common support overlap.

$$p(x) = \max\left(0.15, \min\left(0.85, \frac{1}{1 + \exp(-(-1.5 + 0.15 \cdot \text{PriorSpills} + 0.01 \cdot \text{AssetAge}))}\right)\right)$$

- **True Treatment Effect (τ):** Deeply heterogeneous. Smart sensors yield nominal efficiency savings globally, but generate massive financial returns by preventing catastrophic failures in aging, Combined Sewer Systems (CSS) located in dense Urban Cores.

$$\tau(x) = 5000 + 350 \cdot \text{AssetAge} + \begin{cases} 40000 + 35000 \cdot I(\text{Catchment} = \text{Urban Core}) & \text{if CSS} \\ 15000 \cdot I(\text{Catchment} = \text{Industrial}) & \text{if Separate Sewer} \end{cases}$$

- **Observed Outcome:**

$$Y_0 = 2000 + 50 \cdot \text{AssetAge} + \epsilon_{\text{base}} \quad (\epsilon_{\text{base}} \sim \mathcal{N}(0, 1500^2))$$

$$Y = \max(0, Y_0 + D \cdot \tau(x) + \epsilon)$$

$$\epsilon \sim \mathcal{N}(0, 8000^2)$$

In all the budget_capacity settings and the primary testing focus for each of the ten policy scenarios.

Table 2. In this causal machine learning pipeline, budget_capacity defines the resource constraint under which the policy must operate (i.e., the maximum percentage of the total applicant pool that can receive the treatment).

SCENARIO	BUDGET CAPACITY	PRIMARY TESTING FOCUS
1. AFTER-SCHOOL TUTORING	0.30	Targeted intervention for low-SES/low-score students.
2. REMOTE PATIENT MONITORING	0.20	Triage efficiency for high-risk patients.
3. ENERGY EFFICIENCY RETROFIT	0.15	Prioritising older residential stock for grants.
4. SMALL BUSINESS DIGITAL TRANS.	0.10	Capacity-constrained selection based on business size.
5. SMART IRRIGATION SUBSIDY	0.25	Targeting large, tech-capable, water-stressed farms.
6. ACADEMIC STAFF INCENTIVES	0.25	Pilot program testing for impact-based selection.

7. GENERAL TRAINING INCENTIVES	0.20	Neutral, RCT-style baseline for algorithm validation.
8. URBAN PUBLIC TRANSIT SUBSIDY	0.35	Accessibility optimisation for low-income commuters.
9. URBAN TREE CANOPY MAINTENANCE	0.20	Targeting zones vulnerable to heat island effects.
10. WASTEWATER INFRASTRUCTURE	0.25	Risk-based prioritisation for underground asset repair.

These scenarios are designed to test how a Causal Forest algorithm optimises the allocation of limited resources (defined by budget_capacity) to maximise an outcome metric (Y) while accounting for selection bias and heterogeneous treatment effects.

7. Output of Accompanying Scripts

All ten scripts, namely, causal_forest_academic_staff_incentives.R, causal_forest_afterschool_intensive_tutoring.R, causal_forest_job_training_interventions.R, causal_forest_remote_patient_monitoring.R, causal_forest_residential_energy_efficiency_retrofit.R, causal_forest_small_business_digital_transformation.R, causal_forest_smart_irrigation_subsidy.R, causal_forest_urban_public_transit_subsidy.R, causal_forest_urban_tree_canopy_maintenance.R, and causal_forest_wastewater_policy.R, produce four visualisations, for example, for causal_forest_remote_patient_monitoring.R, the plotted results are:

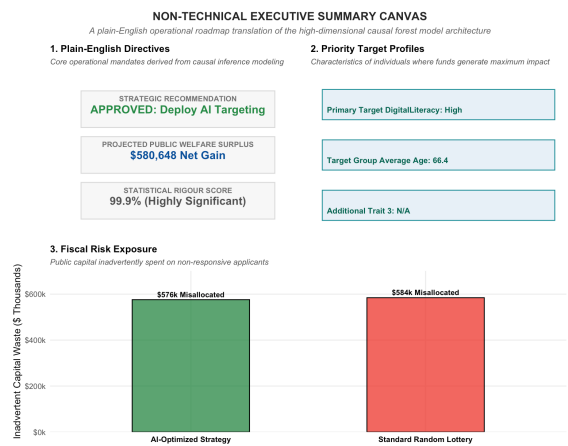


Figure 1: visualisation of executive_summary_canvas object.

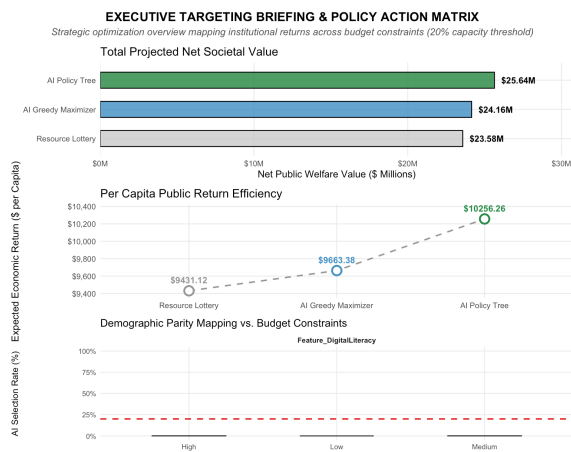


Figure 2: visualisation of policymaker_briefing_dashboard object.

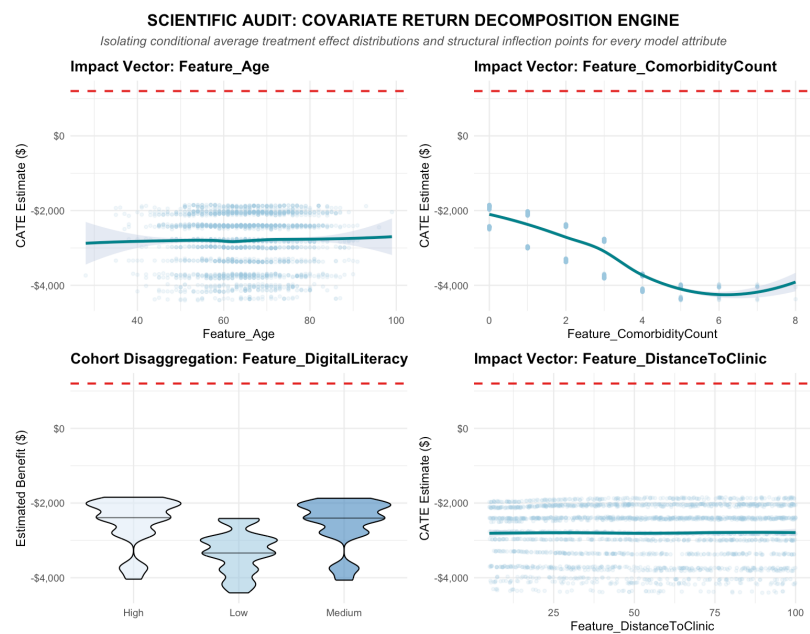


Figure 3: visualisation of covariate_matrix_layout object.

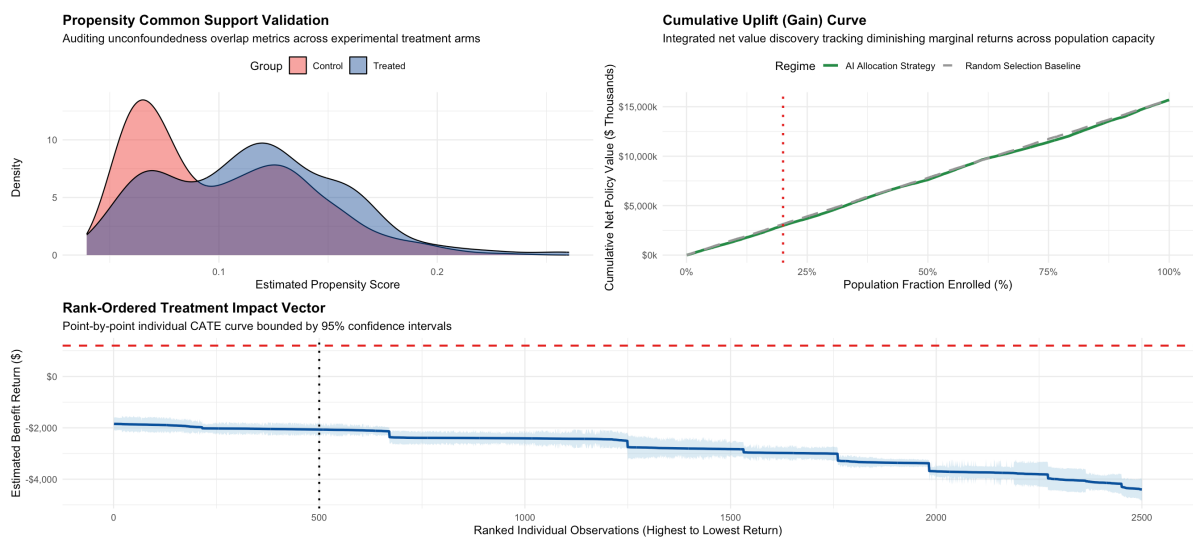


Figure 4: Visualisation of scientific_validation_suite object.

APPENDIX A. Understanding CART and Causal Tree Metrics

The mathematical formulas for the **CART metric** and the **Causal Tree metric** reflect two completely different optimisation objectives. While CART targets the minimisation of variance in the *outcome variable* (Y), the Causal Tree targets the maximisation of variance in the *treatment effect* (τ).

Here is the direct translation of your R code into formal mathematical notation:

A.1. The Standard CART Metric (Variance Reduction)

In the code given at the end, the CART metric is calculated as $\text{var_parent} - ((n_l/N) \cdot \text{var}_l + (n_r/N) \cdot \text{var}_r)$. Mathematically, for a candidate split t , this is expressed as:

$$\Delta_{\text{CART}}(t) = \sigma_{\text{parent}}^2 - \left(\frac{N_L}{N} \sigma_L^2 + \frac{N_R}{N} \sigma_R^2 \right)$$

Component Breakdown:

- σ_{parent}^2 : The total population variance of the outcome Y in the parent node before splitting.
- N_L, N_R : The number of observations that fall into the left and right child nodes, respectively ($N = N_L + N_R$).
- σ_L^2, σ_R^2 : The localised population variance of the outcome Y within the left and right child nodes.

What it optimises:

This formula computes **Mean Squared Error (MSE) minimization**. The algorithm searches for a split that makes the outcome values within each child node as homogeneous (similar to each other) as possible by shedding the maximum amount of variance.

A.2. The Causal Tree Metric (Treatment Effect Heterogeneity)

In the code given at the end, the Causal metric is calculated as $(n_l * n_r / (N^2)) * ((\tau_l - \tau_r)^2)$. Mathematically, for a candidate split t , this is expressed as:

$$\Delta_{\text{Causal}}(t) = \frac{N_L \cdot N_R}{N^2} (\hat{\tau}_L - \hat{\tau}_R)^2$$

Where the treatment effects $(\hat{\tau}_L, \hat{\tau}_R)$ within each node are calculated via the difference-in-means:

$$\hat{\tau}_L = \bar{Y}_L(D = 1) - \bar{Y}_L(D = 0) \quad \text{and} \quad \hat{\tau}_R = \bar{Y}_R(D = 1) - \bar{Y}_R(D = 0)$$

Component Breakdown:

- $\frac{N_L \cdot N_R}{N^2}$: A **sample-size weighting penalty**. It prevents the tree from isolating tiny, unrepresentative outlier clusters (which might show extreme treatment differences purely due to noise) by favouring splits that balance sample distribution.
- $(\hat{\tau}_L - \hat{\tau}_R)^2$: The squared difference between the treatment effects calculated in the left child node vs. the right child node.

What it optimises:

This formula computes **CATE Variance Maximization** (specifically, Athey and Imbens' *EM-Criterion* for causal trees). Because we can never observe the true individual treatment effect (τ_i) for a single person, we cannot minimise MSE directly around τ . Instead, the math exploits a variance decomposition identity: maximising the squared distance between subgroup averages $(\hat{\tau}_L - \hat{\tau}_R)$ is mathematically equivalent to maximising the variance of the predicted treatment effects across the population.

Table A1. Summary of Mathematical Divergence

MATHEMATICAL PROPERTY	STANDARD CART METRIC	CAUSAL TREE METRIC
PRIMARY VARIABLE EVALUATED	Outcome Vector (Y)	Joint Interaction of Outcome, Treatment, and Assignment (Y, D)
MATHEMATICAL OPERATOR	Subtraction (Parent Variance – Child Variances)	Multiplication (Weighted Sample Ratio $\times$ Squared Delta)
OPTIMISATION DIRECTION	Minimises within-node variance of the level of Y .	Maximises between-node variance of the treatment lift (τ).
RESPONSE TO BASELINE SHIFTS	Highly sensitive. Will split immediately on large shifts in Y ,	Completely blind. If treatment response is identical across a split, $(\hat{\tau}_L - \hat{\tau}_R)^2 = 0$, giving a score of zero.

even if treatment response is identical.

Applying **both metrics** on the exact same dataset is the single most effective way to dismantle a dangerous intuition common among data scientists: the belief that *a highly accurate predictive model is automatically a good policy model*. By contrasting the standard CART metric against the Causal metric, you expose how standard machine learning and causal machine learning look at the exact same data matrix through completely different lenses.

Here are the three structural and pedagogical reasons why a rigorous curriculum must leverage both:

Exposing the "Targeting the Sickest" Fallacy (The Policy Pivot)

In public policy, medical triage, and enterprise settings, decision-makers frequently fall into the trap of running a standard predictive model (like CART), finding the population with the lowest projected outcomes (Y), and dumping intervention resources into that bucket.

Running both metrics side-by-side reveals the catastrophic flaw in this approach using the data mechanics from your script:

- **What CART finds:** CART tracks Feature Z because it spots a massive \$50 baseline jump between individuals above and below the threshold ($Z < 50$ vs. $Z \geq 50$). It isolates the "at-risk" or "low-outcome" cohort ($Z < 50$).
- **What Causal Tree finds:** The Causal Tree completely ignores Z because the *lift* (τ) is exactly identical (+30 or +0) on both sides of Z . Instead, it isolates Feature X .

The Lesson: If a policymaker uses the CART metric to allocate a scarce budget, they will target individuals based on their baseline trait Z . However, if those individuals happen to have $X \geq 50$, their treatment return is **exactly \$0**. The budget is entirely wasted on non-responders. The Causal metric shifts the mind from *prognosis* (predicting states) to *responsiveness* (predicting returns).

Building the Foundation for Robinson's Orthogonalisation

Modern, enterprise-grade causal engines (like the grf package) do not actually abandon the CART metric, **they use both metrics sequentially** to isolate unconfounded effects:

1. **The Nuisance Step (CART Engine):** We must first build a predictive model of the baseline outcome surface ($\hat{m}(x) = E[Y|X]$) to account for main effects. This is exactly what the CART variance-reduction metric does.
2. **The Residual Cleanroom:** We subtract that predicted baseline from the actual outcome ($Y - \hat{m}(x)$) to get clean residuals.
3. **The Causal Step (Causal Engine):** We then run the Causal Tree metric on those residuals to map out pure, unconfounded treatment effects (τ) without baseline interference.

Without a firm grasp of the standard CART metric, students view the baseline de-biasing steps of modern causal forests as a black box rather than a logical progression.

The Dual Real-World Decision Matrix

In a clinical or economic deployment, an optimal decision engine requires **both metrics** to function ethically and efficiently.

Consider how an administrator must combine the outputs of both metrics to tier a population:

HIGH RESPONSIVENESS	LOW RESPONSIVENESS
(High Causal Tree Score)	(Low Causal Tree Score)

	CRITICAL TARGET	THE INHERENTLY ROBUST
HIGH BASELINE OUTCOME (Low CART Urgency)	High baseline, but can still be optimized further by the policy.	Doing great on their own; zero reason to intervene.
	MAXIMUM PRIORITY	THE INTRACTABLE CASE
LOW BASELINE OUTCOME (High CART Urgency)	In severe distress AND highly responsive to the intervention.	In severe distress, but completely non-responsive to this specific policy.

- **If you only use CART:** You treat the entire bottom row ("Low Baseline Outcome"), wasting money on the bottom-right quadrant where the treatment has no effect.
- **If you only use the Causal Tree:** You treat the entire left column ("High Responsiveness"), treating affluent or low-risk individuals in the top-left quadrant who might not require public assistance or urgent clinical priority.

A.3. Example R code justifying above concepts:

```
# Tutorial 1: CART and Causal Metric Functions

set.seed(42)
N <- 1000

# 1. Synthesize data streams
X <- runif(N, 0, 100) # Treatment Lift Driver
Z <- runif(N, 0, 100) # Main Effect Driver
D <- rbinom(N, 1, 0.5) # Random Treatment Assignment

baseline <- ifelse(Z < 50, 10, 60)
true_tau <- ifelse(X < 50, 30, 0)
Y <- baseline + D * true_tau + rnorm(N, mean=0, sd=2)

df <- data.frame(X=X, Z=Z, D=D, Y=Y)
var_parent <- var(Y) * (N-1)/N # Population variance helper

# ---- Scoring Functions ----
get_cart_metric <- function(feature,t) {
  left <- df[df[[feature]] < t, ]
  right <- df[df[[feature]] >= t, ]
  n_l <- nrow(left); n_r <- nrow(right)
  var_l <- var(left$Y)*(n_l-1)/n_l; var_r <- var(right$Y)*(n_r-1)/n_r
  return(var_parent - ((n_l/N)*var_l + (n_r/N)*var_r))
}

get_causal_metric <- function(feature,t) {
  left <- df[df[[feature]] < t, ]
  right <- df[df[[feature]] >= t, ]
  n_l <- nrow(left); n_r <- nrow(right)
  tau_l <- mean(left$Y[left$D==1]) - mean(left$Y[left$D==0])
  tau_r <- mean(right$Y[right$D==1]) - mean(right$Y[right$D==0])
  return((n_l * n_r / (N^2)) * ((tau_l - tau_r)^2))
}

# Automated Grid Search Sweep Engine
thresholds <- seq(10, 90, by = 1)

cart_Z_scores <- sapply(thresholds, function(t) get_cart_metric("Z", t))
causal_X_scores <- sapply(thresholds, function(t) get_causal_metric("X", t))

opt_cart <- thresholds[which.max(cart_Z_scores)]
opt_causal <- thresholds[which.max(causal_X_scores)]

# Tutorial 2: Growing the Tree Branches Forward

# Step 1: Physically partition the parent data frame using Layer 1 splits
df_cart_left <- df[df$Z < opt_cart, ]
df_cart_right <- df[df$Z >= opt_cart, ]

df_causal_left <- df[df$X < opt_causal, ]
```

```

df_causal_right <- df[df$X >= opt_causal, ]

# Step 2: Define localized Layer 2 scoring functions bound to child data
get_cart_metric_l2 <- function(sub_df, feature, t) {
  n_sub <- nrow(sub_df)
  if (n_sub < 20) return(0) # Stopping rule: skip if parent subset is too small

  # Total population variance helper localized to this node
  var_sub <- var(sub_df$Y) * (n_sub - 1) / n_sub

  left <- sub_df[sub_df[[feature]] < t, ]
  right <- sub_df[sub_df[[feature]] >= t, ]
  n_l <- nrow(left); n_r <- nrow(right)

  if (n_l < 10 || n_r < 10) return(0) # Stopping rule: reject tiny child nodes

  var_l <- var(left$Y) * (n_l - 1) / n_l
  var_r <- var(right$Y) * (n_r - 1) / n_r
  return(var_sub - ((n_l / n_sub) * var_l + (n_r / n_sub) * var_r))
}

get_causal_metric_l2 <- function(sub_df, feature, t) {
  n_sub <- nrow(sub_df)
  left <- sub_df[sub_df[[feature]] < t, ]
  right <- sub_df[sub_df[[feature]] >= t, ]
  n_l <- nrow(left); n_r <- nrow(right)

  if (n_l < 10 || n_r < 10) return(0) # Stopping rule

  # Causal Stratification Check: Ensure treated and controls exist in both proposed paths
  if (sum(left$D == 1) < 2 || sum(left$D == 0) < 2 ||
      sum(right$D == 1) < 2 || sum(right$D == 0) < 2) return(0)

  tau_l <- mean(left$Y[left$D == 1]) - mean(left$Y[left$D == 0])
  tau_r <- mean(right$Y[right$D == 1]) - mean(right$Y[right$D == 0])
  return((n_l * n_r / (n_sub^2)) * ((tau_l - tau_r)^2))
}

# Step 3: Run Layer 2 optimization sweeps inside the Left Child Nodes
cart_l2_Z_score <- max(sapply(thresholds, function(t) get_cart_metric_l2(df_cart_left, "Z",
t)))
cart_l2_X_score <- max(sapply(thresholds, function(t) get_cart_metric_l2(df_cart_left, "X",
t)))

causal_l2_Z_score <- max(sapply(thresholds, function(t) get_causal_metric_l2(df_causal_left,
"Z", t)))
causal_l2_X_score <- max(sapply(thresholds, function(t) get_causal_metric_l2(df_causal_left,
"X", t)))

# Step 4: Display Structural Diagnostic Insights
cat("\n--- LAYER 2 OPTIMIZATION INSIGHTS ---\n")
cat(sprintf("CART Left Branch Split Options -> Peak Score on Z: %.2f | Peak Score on X: %.2f\n",
            cart_l2_Z_score, cart_l2_X_score))
cat(sprintf("Causal Left Branch Split Options -> Peak Score on Z: %.2f | Peak Score on X:
            %.2f\n",
            causal_l2_Z_score, causal_l2_X_score))

# Tutorial 3: Breaking Causal Trees with Confounding Bias

set.seed(42)
N <- 1000

X <- runif(N, 0, 100)
Z <- runif(N, 0, 100)

# Inject Confounding: High Z individuals are highly likely to get treated
prob_treatment <- ifelse(Z < 50, 0.1, 0.8)
D <- rbinom(N, 1, prob_treatment)

baseline <- ifelse(Z < 50, 10, 60) # High Z has a huge baseline outcome shift
true_tau <- ifelse(X < 50, 30, 0) # X remains the true treatment driver
Y <- baseline + D * true_tau + rnorm(N, mean=0, sd=2)

df_confounded <- data.frame(X=X, Z=Z, D=D, Y=Y)

# Evaluate a split on Feature Z using the simple difference-in-means metric

```

```

left_Z <- df_confounded[df_confounded$Z < 50, ]
right_Z <- df_confounded[df_confounded$Z >= 50, ]

tau_hat_left_Z <- mean(left_Z$Y[left_Z$D==1]) - mean(left_Z$Y[left_Z$D==0])
tau_hat_right_Z <- mean(right_Z$Y[right_Z$D==1]) - mean(right_Z$Y[right_Z$D==0])

cat("\n--- THE CONFOUNDING CRISIS ---\n")
cat(sprintf("Hallucinated Leaf Effect on Low Z: %.2f (True Effect is uniform across Z)\n",
tau_hat_left_Z))
cat(sprintf("Hallucinated Leaf Effect on High Z: %.2f\n", tau_hat_right_Z))

# Robinson's Orthogonalisation (Residualization)

# 1. Regress Y on Z to strip out baseline outcome shifts
m_x <- lm(Y ~ Z, data=df_confounded)
Y_tilde <- residuals(m_x)

# 2. Regress D on Z to strip out propensity selection bias
e_x <- glm(D ~ Z, data=df_confounded, family=binomial)
D_tilde <- residuals(e_x)

# 3. Evaluate the true driver (X) using the unconfounded residuals
df_residualized <- data.frame(X=X, Y_tilde=Y_tilde, D_tilde=D_tilde)

```

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- Robinson, P. M. (1988). Root-N-consistent semiparametric regression. *Econometrica*, 56(4), 931–954.
- Wager, S., & Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523), 1228–1242.